

Differential Diagnosis

Treatment

Pharyngotonsillar diphtheria requires treatment with antibiotics (to kill the bacteria) and antitoxin (to neutralize the exotoxin). Treatment should be started on clinical suspicion without awaiting culture confirmation, because toxin produced in the interim might damage the myocardium irreversibly.

Cutaneous diphtheria must be treated with antibiotics and antitoxin (see Table 183-1). The antibiotic of choice is high-dose intravenous penicillin or, for those unable to take penicillin, erythromycin. Diphtheria antitoxin, available from the Centers for Disease Control and Prevention, is administered intravenously after careful testing for horse serum hypersensitivity. Injection of antitoxin subcutaneously around and under the ulcer, along with superficial application to the lesion, may be beneficial, but data are incomplete. After antitoxin and antibiotics are administered, the membrane should be debrided and the ulcer kept clean. Patients with either form of the disease should be isolated for roughly 14 days.

Prognosis, Clinical Course, and Complications

Prevention

OTHER CUTANEOUS DIPHTHEROID INFECTION

CHAPTER 184

Tuberculosis and Infections with Atypical Mycobacteria

Gerhard Tappeiner

The genus *Mycobacterium* is estimated to cause more suffering for humans than all other bacterial genera combined. Twenty-five years ago, it was assumed that tuberculosis would become extinct in the developed countries, because its incidence decreased at an average rate of 6 percent in the United States and 10 percent in Europe between 1953 and 1985. However, by 1983, tuberculosis was declared a "global emergency" by the World Health Organization due to a sharp increase in incidence elsewhere (e.g., a doubling in sub-Saharan Africa), largely driven by the human immunodeficiency virus (HIV) pandemic, poverty, and social disruption. One-third of the world's population is thought to be infected. In the United States, the figure is estimated at 15 million. Currently,

an estimated 8 million new cases of tuberculosis and 3 million tuberculosis-related deaths occur each year, 98 percent of them in developing countries. There are also 1.5 million new cases and 500,000 deaths per year among children. Tuberculosis has the highest morbidity and mortality rates of all infectious diseases and is thought to be responsible for 25 percent to 30 percent of avoidable deaths.

The so-called atypical mycobacteria (mycobacteria other than *M. tuberculosis*, or MOTT) cause skin disease more frequently than does *M. tuberculosis*. They exist in various reservoirs in the environment. Among these organisms are obligate and facultative pathogens as well as non-pathogens. In contrast,

KEY REFERENCES

The full reference list for all chapters is available at www.digm7.com.

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2. Hayes EB: Tularemia, in *Tick-Borne Diseases of Humans*, edited by Goodman JL, Dennis DT, Sonenshine DE. Washington, DC, ASM Press, 2005, p 207
3. Pappas G et al: Brucellosis